# Timofey Biryukov

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## EDUCATION

**B.S. Mechanical Engineering** May 2027

*Virginia Tech, Blacksburg, VA*

- Relevant coursework: Fluid/Thermal Dynamics, Heat and Mass Transfer, Mechanical Design
- Design team lead on Naval Surface Warfare Center funded Senior Design project

## SKILLS

- Microsoft Suite
- Adobe Suite
- MATLAB/Python
- Problem Solving
- 3D Printing
- Waveform Analysis
- ANSYS Fluent
- Technical Communication
- CNC Programming
- Proficiency in CAD (Inventor, Fusion, Solidworks)

## WORK EXPERIENCE

**Undergraduate Research Assistant** Feb 2025 - Present

*Virginia Tech, Blacksburg*

*Dr. Guo-Quan Lu's Power Electronics Materials, Modules, and Manufacturing Laboratory*

- **Designed and experimentally validated** a water jet-impingement cooling system for an Office of Naval Research sponsored project, using CFD simulations to optimize thermal and fluid flow performance under high heat flux conditions.
- **Engineered experimental two-phase cooling system** to characterize phase-change heat transfer limits and mitigate critical heat flux bottlenecks.
- **Characterized SiC power module packaging** with Transient Dual Interface Method testing in accordance with JEDEC standards, quantifying the junction-to-case thermal impedance.
- **Conducted thermal characterization and reliability testing** for wide-bandgap Silicon Carbide power electronic modules.
- **Contributed to multiple peer-reviewed research papers** on advanced packaging and electronics cooling, with a primary authorship digest underway for the 2027 IEEE Energy Conversion Congress and Exposition.
- **Directed** a STEM mentorship initiative, supervising high school and other undergraduate researchers in testbench design, data acquisition, and thermal engineering fundamentals.

**Electronics Advisor** Aug 2022 - Aug 2024

*Best Buy, Vienna, VA*

- Provided technical support and basic computer repairs for retail customers, troubleshooting hardware and software issues.
- Exceeded $100k in monthly retail sales and successfully drove credit card enrollment targets.

**Product Content Specialist** Oct 2022 - Oct 2023

*Mervis Diamond Importers, Tysons, VA*

- Photographed and photoshopped high end jewelry for optimal consumer appeal.
- Managed online store presence and listings via Shopify and Excel spreadsheets.

**Small Business Owner** June 2022 - Present

*Tomorrow Devices, Vienna, VA*

- Produced specialized electronic equipment for local businesses, focusing on power-washing facilitation devices.
- Designed and assembled RF circuits, including component sourcing and end-to-end product development.
- Managed marketing efforts and customer relationships to drive satisfaction and repeat business.